

Generation Date: 01-Dec-2024 04:53:20 (applicant copy)

THE UNIVERSITY OF HONG KONG

Application for Admission to the Doctor of Philosophy (PhD) and
Master of Philosophy (MPhil) ProgrammesRound of Admission: Main Round for 2025–26 Admission
(username: zhouxiangru90)

1. Programme Apply For

Programme:	PhD
Study mode:	Full-time
Proposed faculty /Interdisciplinary area:	Interdisciplinary Area
Proposed department/area of study:	Biomedical Engineering
Proposed discipline:	
Proposed field of study:	Brain Computer Interface
Proposed supervisor(s):	
Proposed registration date: (DD/MM/YYYY)	1/9/2025
Apply for joint PhD programme:	No
Partner institution of joint PhD programme:	
Apply for work-study programme:	No
Partner Institute of work-study programme:	
Apply for PGS:	Yes
Apply for HKPFS:	No
HKPFS reference number:	

2. Personal Information

Title:	Mr.
Surname:	ZHOU
Given name:	Xiangru
Middle name:	
Full name in Chinese:	周相如
Sex:	M
Date of birth (DD/MM/YYYY)	31/7/1990
HKID card number:	F601647(9)
Passport number:	
Chinese National ID:	450881199007318019
HKU student number:	
Country of residence:	China
Nationality (Place of origin):	China
Correspondence address:	Room 2205, Building 12, Qinghaian Garden, Nansha Street, Nansha District, Guangzhou, Guangdong, China.

Postal code:	511458
Email address:	(1) xiangru.zhou@connect.polyu.hk (2) JohnsonZhouXR@outlook.com
Contact phone number:	(1) +8618502091962 (2) 85264820616
Right of abode:	N
Need visa/entry permit:	Y
Type of visa/entry permit:	Student visa/entry permit
First issued date of visa/entry permit (DD/MM/YYYY):	

3. Academic Qualification

3.1 Current Academic Studies

[Only one record is allowed. Applicants with more than one current academic studies are advised to provide the complete information in Section 7 Additional Information.]

(1)	Level of qualification:	Other Master's degree (Full-time)
	Awarding institution:	Hong Kong Polytechnic University (The)
	Awarding country:	Hong Kong(SAR)of PRC
	Major subject:	Microelectronics Technology & Materials
	Start date of attendance:	9/2024
	Year(s) of study completed:	1
	Expected date of award (DD/MM/YYYY):	1/7/2025
	Cumulative grade point average (CGPA):	
	Average mark:	
	Language of instruction:	English
	Qualification obtained by:	By Coursework

3.2 Previous Academic Studies

(1)	Level of qualification:	Bachelor's Degree (Full-time)
	Awarding institution:	Xi'an University of Technology
	Awarding country:	China
	Major subject:	Microelectronics
	Date of attendance:	From 9/2009 to 7/2013
	Year(s) of study completed:	4
	Status:	Graduated
	Title of award:	Bachelor of Engineering
	Date of award (DD/MM/YYYY):	1/7/2013
	Honours classification:	Not Applicable
	Cumulative grade point average (CGPA):	
	Average mark:	
	Language of instruction:	Not English
	Qualification obtained by:	By Coursework

4. Professional Qualification, Prize/Award and Work Experience

4.1 Professional Qualification, Prize and Award

Nil

4.2 Work Experience

	Name of organisation	Position	Mode	Employment period (DD/MM/YYYY)	Duration of employment (in days)	Major responsibilities
(1)	SmartMore Corporation Limited	Team Leader	Full-time	21/5/2020 to 1/9/2024	1565	Leader of a team of four, C++ & python coder and total solution provider. Key Projects: AI-driven robotic arm Wafer ID Reader Wafer OCR Code Reader Integrated Machine Project Defect Detection System for SONY Labels Character Recognition System for Apple Watch Bands Using Laser Engraving Technology
(2)	Shenzhen PHDI Corporation Limited	Co-founder, CTO	Full-time	2/7/2016 to 20/5/2020	1419	Co-founder, CTO Key Projects: Android Software for A Car-Mounted Mobile Phone Controller Voice Training and Recognition Program Based on The Kaldi Framework Face Recognition and Tracking Software on The Android Platform
(3)	OPT Machine Vision Tech Co., Ltd.	Image-processing Algorithm Engineer	Full-time	25/4/2014 to 1/7/2016	799	Image processing algorithm engineer Key Projects: One-Dimensional Barcode Localization and Recognition Algorithm Two-Dimensional Barcode Localization and Recognition Algorithm Rapid Edge And Circle Detection Algorithm

5. English Language and Other Test Results

5.1 English Language Test

English language test	Date of test (MM/YYYY)	Overall score/grade	Appointment number
TOEFL:			
IELTS (Academic module):	3/2024	6.5	-
GCE:			-
IGCSE:			-
CPE:			-

5.2 Other Tests

Nil

6. Referees

(1)	Full name:	Prof. Jun <u>YIN</u>
	Position:	Associate Professor

Organisation and department:	The Hong Kong Polytechnic University
Personal webpage:	
Relationship to applicant:	Lecturer
Email:	jun.yin@polyu.edu.hk
Contact phone number:	+852 2766 5684
Postal address:	CD604
	11 Yuk Choi Road, Hung Hom,
	KLN, Hong Kong.

(2)	Full name:	Dr. Ruiyu <u>LI</u>
	Position:	Department supervisor
	Organisation and department:	SmartMore Corporation Limited
	Personal webpage:	
	Relationship to applicant:	Former supervisor
	Email:	ryli@smartmore.com
	Contact phone number:	86 13058014285
	Postal address:	22F, Building T2, Qianhai Kerry Center,
		Nanshan street, Nanshan District,
		Shenzhen City, Guangdong Province, China.

7. Additional information

I will upload my research proposal and personal CV in the Section 9.
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学士学位证书

周相如，男，1990年7月31日生。在西安理工大学
微电子学
专业完成了本科学习计划，业已
毕业，经审核符合《中华人民共和国学位条例》的规定，授予工学
学士学位。

西安理工大学

校长

学位评定委员会主席

证书编号：1070042013001855

二〇一三年七月三日

(普通高等教育本科毕业生)





学生学位证明

姓名	周相如
性别	男
出生日期	1990-07-31
入学时间	2009-09-10
毕业时间	2013-07-03
系别	自动化与信息工程学院
专业	微电子学

经审核符合《中华人民共和国学位条例》规定，授予工学学士学位。

学位委员会主席：刘丁

学校(盖章):



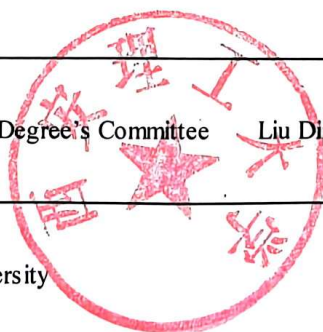
XI'AN UNIVERSITY OF TECHNOLOGY
Bachelor's Degree Certificate

Name	Zhou Xiangru
Gender	Male
Date of Birth	1990-07-31
Data of Enrollment	2009-09-10
Date of Graduation	2013-07-03
Department	Faculty of Automation and Information Engineering
Speciality	Microelectronics

In accordance with“ The Regulation Concerning Academic Degrees in the
People's Republic of China” , the student is awarded the Bachelor of Engineering.

Chairman of Degree's Committee Liu Ding

Seal of University





打印日期:2024 年6月4日

[illegible]

Date: Jun. 4, 2024



西安理工大学
XI'AN UNIVERSITY OF TECHNOLOGY

学生学业成绩表

学院: 自动化与信息工程学院
姓名: 周相如

专业: 微电子学
学号: 3090433019

班级: 微电091

课程名称	课程性质	学分	成绩	绩点	课程名称	课程性质	学分	成绩	绩点
2009-2010学年 第1学期					工程技术综合实践	校级选修课	2.0	优	4.5
英语A(2)	必修课	4.5	86	4.1	2011-2012学年 第1学期				
中国近现代史纲要	必修课	1.0	62	0	数字电子技术	必修课	4.0	91	4.6
高等数学A(上)	必修课	6.0	73	2.8	数字电子技术实验	必修课	0.5	良	3.5
大学计算机基础A	校级选修课	2.5	100	5	数字电子技术课程设计	必修课	1.0	优	4.5
体育(1)	必修课	1.0	80	3.5	固体物理	必修课	3.5	83	3.8
思想道德修养与法律基础	必修课	2.0	61	1.6	半导体物理	必修课	4.5	84	3.9
军训	必修课	0.5	85	4	生产实习1(微电)	必修课	2.0	良	3.5
2009-2010学年 第2学期					半导体基础实验	必修课	1.0	良	3.5
C语言程序设计	必修课	3.5	94	4.9	高频电子线路D	必修课	2.5	76	3.1
C语言程序设计课程设计	必修课	1.0	中	2.5	计算机网络与通信B	院级选修课	2.5	84	3.9
电子技术查新训练	必修课	1.0	优	4.5	体育(3)	必修课	1.0	79	3.4
英语A(3)	必修课	4.5	80	3.5	2011-2012学年 第2学期				
马克思主义基本原理	必修课	2.0	90	4.5	微机原理及应用A	院级选修课	4.0	86	4.1
高等数学A(下)	必修课	6.0	77	3.2	集成电路工艺原理	必修课	3.5	94	4.9
线性代数	必修课	2.5	84	3.9	微电子技术基础	必修课	4.0	83	3.8
物理实验(上)	必修课	1.0	良	3.5	生产实习2(微电)	必修课	2.0	优	4.5
大学物理A(上)	必修课	3.5	84	3.9	微电子技术综合实践	必修课	2.0	良	3.5
图学基础及CAD	必修课	3.0	86	4.1	半导体材料	院级选修课	2.0	71	2.6
高分子世界概论	校级选修课	2.5	优	4.5	MCS51单片机原理及应用	院级选修课	2.0	82	3.7
体育(2)	必修课	1.0	85	4	光电子学原理	院级选修课	2.5	90	4.5
军事理论	必修课	0.5	88	4.3	体育(4)	必修课	1.0	90	4.5
艺术导论	校级选修课	1.5	66	2.1	2012-2013学年 第1学期				
2010-2011学年 第1学期					半导体集成电路A	必修课	4.0	86	4.1
计算机软件基础	必修课	3.5	84	3.9	集成电路设计技术A	必修课	3.0	76	3.1
电路(1)	必修课	4.0	80	3.5	半导体专业实验	必修课	1.5	优	4.5
电路(1)实验	必修课	0.5	中	2.5	集成电路设计实践	必修课	2.0	良	3.5
英语A(4)	必修课	4.5	74	2.9	检测与传感技术(电子、微电)	院级选修课	2.0	81	3.6
日语(中级)	校级选修课	2.5	90	4.5	射频集成电路设计技术	院级选修课	2.5	86	4.1
毛泽东思想和中国特色社会主义理论体系概论	必修课	3.0	69	2.4	电力半导体器件	院级选修课	2.0	73	2.8
复变函数与积分变换	必修课	2.5	61	1.6	2012-2013学年 第2学期				
概率论及数理统计B	必修课	3.0	87	4.2	毕业设计(微电)	必修课	17.0	良	3.5
物理实验(下)	必修课	1.0	优	4.5	以下空白				
大学物理A(下)	必修课	4.0	78	3.3					
工程训练 A	必修课	3.0	良	3.5					
2010-2011学年 第2学期									
新型复合材料	校级选修课	1.5	89	4.4					
模拟电子技术	必修课	4.0	82	3.7					
模拟电子技术实验	必修课	0.5	中	2.5					
理论物理基础	必修课	3.5	61	1.6					
数理方程	必修课	2.0	87	4.2					
电子线路CAD综合训练	必修课	2.0	良	3.5					
中国传统文化	校级选修课	2.0	90	4.5					
英语A(5)	必修课	4.5	80	3.5					
已获得总学分:184.0 必修课:150.0 校级选修课:14.5 院级选修课:19.5									
平均学分绩点:3.64 加权平均成绩:71.8 诚信:466; 4.5									

审核:

张冠华

院系(盖章)

自动化与信息
工程学院

教务处盖章:



打印日期: 2024-05-30

Advancing Brain-Computer Interface (BCI) Technology

Background and Motivation

I am currently a Master's student in Microelectronics Technology and Materials at The Hong Kong Polytechnic University. Before pursuing my Master's degree, I accumulated over ten years of professional experience in developing image processing algorithms and AI-based image recognition solutions. I previously worked as an Algorithm Engineer and R&D Lead at notable organizations, including the publicly listed company OPT and the prominent AI firm SmartMore Technology.

My skill set includes programming, engineering debugging, setting up and deploying integrated development environments, data collection and preprocessing, parameter optimization, model training and testing, result analysis and refinement, as well as replicating and enhancing cutting-edge technologies. I have successfully filed and been granted more than ten invention patents. Additionally, I possess expertise in designing Bluetooth communication devices, drawing circuit schematics, and creating 3D models and printed components. My prior projects include using AI technology to control multi-axis robotic arms and other complex systems.

I have long been interested in Brain-Computer Interface (BCI) technology, which I regard as a guiding vision for my career development. To me, BCI is a multidisciplinary field that requires a balanced focus: 50% on engineering and computer science (including AI and electronics), 20% on materials science and microphysics, and 30% on bioengineering and medical science (especially neuroscience and brain science). Over the past decade, I have concentrated on software development, image processing, and AI applications while maintaining a strong passion for computer and electronic technologies. With the rapid advancements in BCI, I believe now is the perfect time to take a decisive step forward into this field.

Current Focus and Future Plans

To transition into BCI research, I have revisited foundational technologies, such as cochlear implants and OpenBCI systems, while reorganizing my technical expertise. With a solid background in image processing, software development (mainly in C++ and Python), and AI applications, I decided to pursue a Master's degree in Semiconductor Materials and Technology to address my knowledge gaps in nanomaterials and microphysics. These interdisciplinary skills are critical for improving the precision and accuracy of signal acquisition in BCI, where electrode materials and their interactions with biological signals are particularly important.

Looking ahead, I plan to complement my engineering expertise with foundational knowledge in biology, neuroscience, and medical science, enabling me to collaborate more effectively with experts in BCI teams. My current research focus will continue to center on software development, signal acquisition and processing, and model training and optimization. By analyzing large datasets of neural signals, I aim to identify patterns and use AI techniques to enhance signal recognition accuracy. Moreover, I am particularly interested in exploring how brain signals can control robotic arms, combining my prior experience to develop better rehabilitation solutions for patients.

Research Goals

If given the opportunity to pursue a PhD, I aim to achieve the following research objectives:

- **Data Acquisition and Analysis:** Collect and preprocess large volumes of neural signal data, ensuring its quality through meticulous cleaning, classification, and validation.
- **Model Development:** Design and refine brain signal recognition models, leveraging AI technology to improve accuracy, robustness, and stability.
- **Real-World Applications:** Investigate the feasibility of brain-controlled devices, particularly in integrating BCI technology with robotics.
- **Interdisciplinary Collaboration:** Combine software engineering with neuroscience, brain science, and materials science to deliver innovative BCI solutions, especially focusing on advancements in electrode technology.

Through these efforts, I aspire to contribute to the development and clinical validation of BCI technologies, ultimately making a meaningful societal impact.

Challenges and Preparedness

I fully recognize the challenges in this field, such as low initial recognition rates for models, stability issues, and difficulties in transferring results across devices or environments. However, my extensive professional experience equips me well to address these challenges. I understand the critical role of data quality in determining the success of a model and how improvements in sensor precision, signal processing techniques, and experimental setups can significantly enhance research outcomes.

Furthermore, I am aware of the importance of interdisciplinary collaboration. For instance, in BCI, electrode materials play a decisive role in signal acquisition accuracy and reliability, and understanding their properties and interactions with biological systems is crucial. Similarly, collaborating with medical professionals requires a solid grasp of medical protocols and effective communication. By equipping myself with

knowledge in neuroscience, materials science, and engineering, I am preparing methodically to overcome these challenges.

Commitment to BCI and the Greater Bay Area

I have deep ties to the Greater Bay Area, with my family based in Nansha District, Guangzhou, and eight years of professional experience in Shenzhen, including four years as an entrepreneur. I am fluent in Cantonese, Mandarin, and English (with basic proficiency in Japanese), which enables me to thrive in the region's international innovation ecosystem.

The Greater Bay Area provides a unique environment for advancing BCI technology, offering world-class medical facilities, abundant data resources, and a vibrant culture of innovation. I am committed to establishing myself in this region, channeling my research findings back into the local community to drive technological development and enhance social welfare.

Conclusion

Thank you for taking the time to review my research proposal. I am eager to contribute to the advancement of BCI technology and look forward to the opportunity to collaborate with like-minded researchers. Please feel free to contact me if you have any questions or need additional information.

Test Report Form

ACADEMIC

NOTE Admission to undergraduate and post graduate courses should be based on the ACADEMIC Reading and Writing Modules.
GENERAL TRAINING Reading and Writing Modules are **not** designed to test the full range of language skills required for academic purposes.
It is recommended that the candidate's language ability as indicated in this Test Report Form be re-assessed **after two years** from the date of the test.

Centre Number

CN002

Date

10/MAR/2024

Candidate Number

137676

Candidate Details

Family Name

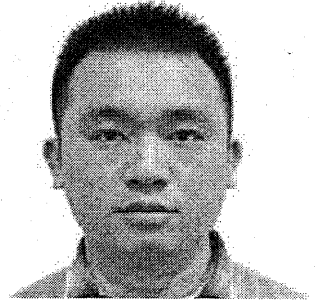
ZHOU

First Name

XIANGRU

Candidate ID

450881199007318019



Date of Birth

31/07/1990

Sex (M/F)

M

Scheme Code

Private Candidate

Country or Region
of Origin

CHINA (PEOPLE'S REPUBLIC OF)

Country of
Nationality

First Language

CHINESE

Test Results

Listening

6.0

Reading

6.5

Writing

6.0

Speaking

7.0

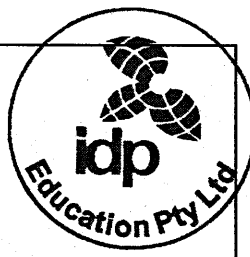
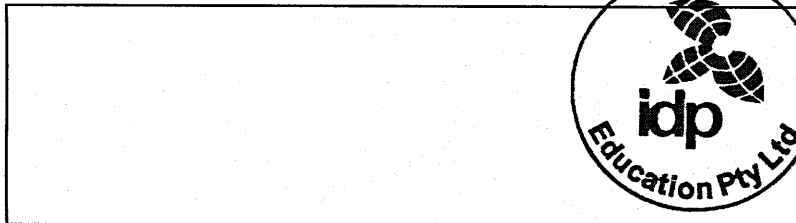
Overall
Band
Score

6.5

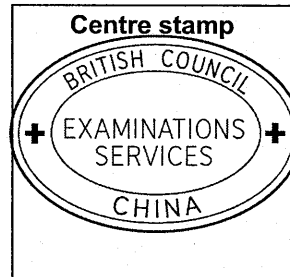
CEFR
Level

B2

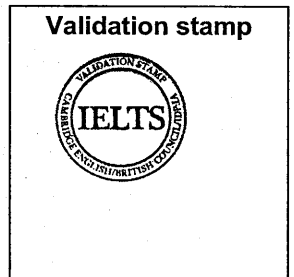
Administrator Comments



Centre stamp



Validation stamp



Administrator's
Signature

Date

13/03/2024

Test Report Form
Number

23CN137676ZH0X002A



Cambridge Assessment
English

Xiangru Zhou

Email: xiangru.zhou@connect.polyu.hk | Telephone: (+86) 18502091962, (852) 64820616

EDUCATION

The Hong Kong Polytechnic University

MSc Microelectronics Technology & Materials

Hong Kong

Sep. 2024 – Present

Xi'an University of Technology

Bachelor of Microelectronics

Xi'an, China

Sep. 2009 – Jun. 2012

• **GPA: 3.64/5.0**

• **Average Score:** 81.18/100

• **Honours:** 2nd Prize 8th Xi'an High-tech "Challenge Cup" Shaanxi Province University Student Extracurricular Academic Science and Technology Works Competition; 1st and 3rd Prize University 19th Extracurricular Academic Science and Technology Works Competition; Special Prize & 2nd Prize University 19th "LiAo Cup"; 1st Prize University 18th "LiAo Cup"; University Innovation Achievement 2nd Prize, University 3rd Prize Scholarship

PROFESSIONAL EXPERIENCE

SMARTMORE CORPORATION LIMITED

Software Engineer

Shenzhen, China

May 2020 – Sep. 2024

AI-Driven Robotic Arm Project

Leader

May 2024 – Sep. 2024

• Branch 1: Traditional Hand-Eye Calibration + Image Recognition

- Implemented 3D scenarios: arbitrary object recognition and grasping, sketching based on verbal commands, and interactive stacking toys with users.
- Integrated an Intel Realsense camera, a structured light 3D camera, a 7-axis robotic arm with a gripper, two servers for large-model deployment, and a PC for control and ASR-based voice input.
- Deployed conversational and vision models to enable seamless interaction between visual, auditory, and motor systems.
- Achieved high precision in object localization and vision-guided robotic operations.

• Branch 2: Transformer-Based Action Prediction

- Utilized a Transformer model to predict robotic arm actions by joint data, CNN features, and task descriptions.
- Enabled real-time responses without hand-eye calibration, adapting to dynamic environments.
- Tackled computational challenges and dataset requirements for effective training and deployment.

Wafer ID Reader Wafer OCR Code Reader Integrated Machine Project

Leader

Oct. 2021 – Sep. 2024

- The first deep learning-based OCR wafer character recognition tool in the semiconductor industry. Convenient use and no need to adjust the recognition parameters compare to its competing products.
- Collaborated with the algorithm team to devise strategies, conduct experiments, and select optimal algorithmic solutions.
- Modified software and SDK interfaces and functionalities based on the product manager's research and user requirements.
- Implemented end-to-end closed-loop process from requirement analysis to deployment, including evaluation, experimentation, algorithm/software/SDK development, testing, and deployment.
- Same recognition rate level as that of industry leader Cognex products in normal cases, partially exceeds its recognition rate (100% versus 99.5%) in the case of fixed format recognition.

Defect Detection System for SONY Labels

Second-hand Collaborator

Feb. 2023 – May 2023

- Conducted Proof of Concept (POC) using client samples to validate system performance.
- By combining the template image and the image to be checked together to form a two-channel image, defects are marked and trained on this basis, the accuracy has been improved by 2%.

BGI Genomics Reagent Bottle Testing

Collaborator

Oct. 2022 – Feb. 2023

- Designed algorithmic solutions and implemented the AI model into software for real-time monitoring of reagent bottle caps, ensuring seal integrity and smoothness during the manufacturing process.
- Conducted Optical Character Recognition (OCR) for character detection on reagent bottle bodies, effectively identifying characters on various colored backgrounds of medication and detecting printing defects.

Specific String OCR System for Arbitrary Orientations On Tags

Designer

Jun. 2021 – Feb. 2022

- Designed the composition of models responsible for the task, leveraging the existing MobileNet training framework in PyTorch. Trained four models individually: label localization, orientation determination, character localization, and recognition.
- Developed a software development kit (SDK) in Visual Studio to facilitate integration and usage of the OCR system.
- Designed and implemented a user-friendly interface using the QT framework to enhance the usability and accessibility of the OCR system.

Character Recognition System for Apple Watch Bands Using Laser Engraving Technology

Designer

May. 2020 – May 2021

- Trained character localization and classification models on a four-card 2080Ti server running Ubuntu, utilizing the MobileNet architecture.
- Employed the ONNX Runtime (ORT) for model deployment, integrating a C++ SDK for inference on the server and developing a user-friendly software interface.
- Collaborated with colleagues to align and optimize the SDK, including pre-processing and post-processing workflows.
- Precision and recall were above 99.7% respectively.

Defect Detection System for Apple Watch Bezels Using 3D Imaging Technology

Designer

Jun. 2020 – Jun. 2020

- Implemented the conversion of 3D point cloud data into 2D heatmaps for analysis purposes.
- Designed and programmed the user interface for the software application.

SHENZHEN PHDI CORPORATION LIMITED

Shenzhen, China

Co-founder, CTO

Jan. 2017 – May 2020

Android Software for A Car-Mounted Mobile Phone Controller

Designer

Jan. 2017 – May 2020

- Managed PCB production designed Bluetooth chip-related programs and conducted 3D design and printing for the product's appearance.
- Enabled rapid switching between navigation, music, and WeChat applications in a car environment.
- Implemented automatic sending and receiving of WeChat messages.
- Integrated features for one-touch activation of voice navigation and song search.
- Designed and developed a comprehensive product including a mobile application and a Bluetooth controller.

Face Recognition and Tracking Software on The Android Platform

Designer

Jan. 2017 – May 2017

- Utilized OpenCV, Caffe, Idlib, and VGG net deep learning neural network for face localization and gender recognition.
- Integrated BP neural network for gender recognition within the face detection algorithm.
- Applied the developed algorithm in systems for estimating the attention of different gender groups towards billboards and for access control systems, resulting in the acquisition of a patent.
- Adapted the algorithm for integration with security cameras to track familiar/strange faces indoors, aiding in the detection of potential security threats.
- Successfully delivered the developed solution to a renowned domestic security camera company.

OPT MACHINE VISION TECH CO., LTD.

Guangdong, China

Image processing algorithm engineer

Apr. 2014 – Jan. 2017

One-Dimensional Barcode Localization and Recognition Algorithm

Designer

Apr. 2014 – Jan. 2017

- Achieved automatic localization of barcodes in approximately 10ms on a PC equipped with an i3 processor, capable of identifying barcodes in complex images of up to 2 million pixels.

Two-Dimensional Barcode Localization and Recognition Algorithm

Designer

Jan. 2015 – Jan. 2017

- Implemented functionality to locate and recognize QRCode, DataMatrix, and other two-dimensional barcodes.
- Utilized Google's open-source libraries ZXing/Zbar and made necessary modifications for enhanced performance

Rapid Edge And Circle Detection Algorithm

Designer

Jan. 2015 – Jan. 2017

- Utilized non-Hough fitting method to swiftly fit specified parameters of lines and circles in designated directions, implementing the algorithm to effectively recognize lines and circles in blurry edge images of up to 2 million pixels.
- Achieved real-time identification within 10ms on a PC with an i3 processor and 8GB of memory.

ADDITIONAL INFORMATION

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- **Language:** English (Fluent; IELTS: 6.5), Japanese (N5)
 - **Programming Language:** C, C++, Python, Java, Java Script, HTML, CSS, PHP
 - **Development tools:** QT, Visual Studio
 - **Systems:** Windows Desktop, Ubuntu Server
 - **Libraries:** OpenCV, PyTorch, Halcon
 - **IT Tools:** Git, Docker, Anaconda
 - **Other Skills:** gdb/pdb, Keil (STM32), Altium Designer, Rhino, Dreamweaver
 - **Books:** Computer Vision: Algorithms and Applications, Design Patterns: Elements of Reusable Object-Oriented Software
 - **Patents:** CN202311136494.1; CN202310973415.6; CN202310287581.0; CN202310051665.4; CN202211602503.7; CN202211474648.3; CN202211357774.0; CN202210994137.8; CN202230264395.1; CN201911330078.9; CN201110120104; CN201110308514.X